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Aurora Kinase A inhibitor alisertib failed to exert its efficacy on TNBC cells due to consequential enrichment of polyploid giant cancer cells (PGCCs)

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Abstract

Background Kinases are emerging as promising targetable choices for Triple negative breast cancer (TNBC) patients who suffer from lack of targeted therapy. Aurora kinase A (AURKA) inhibitor alisertib although showed promise but failed to impart any ultimate benefit in TNBCs. Considering initial preclinical success of alisertib, this study tried to explore the underlying mechanisms of its subsequent insensitivity in TNBCs.

Methods Breast cancer kinome was screened for kinases with high clinical significance by clinical kinase indexing coupled with studying their differential expression pattern and prognostic impact using available datasets. Relevant kinase inhibitors were assessed for pharmacological/toxicological parameters in silico. Alisertib induced polyploid giant cancer cells (PGCCs) were observed microscopically and flow cytometrically. PGCC viability was assessed by single cell MTT and (BrdU) assay. Epithelial and mesenchymal transition (EMT) and stemness markers were examined by immunofluorescence. PGCC enrichment was additionally checked in stained tissue explant/histocultures. Alisertib insensitive cells were developed by repeated alisertib pulsing as confirmed by MTT assay. Alisertib insensitive mammospheres were targeted with mifepristone.

Results Clinical Kinase Index (CKI) scoring and pan cancer expressional profiling coupled with analysis of prognostic effect revealed AURKA as a targetable prognostically relevant kinase. AURKA-targeting agent alisertib demonstrated acceptable pharmacological/toxicological properties but enriched PGCCs in in vitro and tumour histocultures. PGCCs maintained replicative potency producing viable progenies. They retained expression of EMT and stemness markers; demonstrating single cell clonogenicity. PGCC enrichment led to reduced alisertib sensitivity as evident from enrichment of insensitive mammospheres, targetable by mifepristone.

Conclusion PGCCs contributed to alisertib insensitivity in TNBCs that may be targeted by mifepristone.



Keywords Aurora kinase a, Alisertib-insensitivity, Polyploid giant cancer cells (PGCCs), Mifepristone

1 Introduction

Breast cancer cases had risen significantly in recent years making it the most common cancer type among women globally [1]. Among the different molecular subtypes, triple negative breast cancer (TNBC) constitutes 15–20% of all invasive breast cancer cases [2]. TNBCs are characterized by an inherent aggressiveness and enhanced metastatic propensity that culminates in rapid disease progression in such patients. TNBCs lack the expression of the classical steroid hormone receptors (SHRs) like Estrogen Receptor (ER) and Progesterone Receptor (PR) as well as of the Human epidermal growth factor receptor (Her2). Accordingly, targeted therapy directed against these molecular markers are considered to be promising for non-TNBCs but unattainable in TNBCs. Lack of targeted therapy and unavoidable toxicity associated with non-targeted chemotherapy are the major hindrances accounting for poor prognosis of TNBC patients [3]. Thus, looking for alternative targetable markers for TNBCs is a pressing need.

Kinases come up as one of the most promising candidates owing to their major role in carcinogenesis, cell survival, migration as well as cancer cell metastasis [4]. Moreover, kinases represent one of the most targeted group of proteins during drug development for which several kinase inhibitors had been developed so far [5]. Alisertib, being such a kinase inhibitor targeting Aurora Kinase A had shown initial promise in cases of breast cancer clinical trials. Aurora Kinase A is a cell cycle kinase known to be overexpressed in TNBCs and plays a crucial role in dictating cell survival in face of therapy challenge, promotes epithelial to mesenchymal transition (EMT) as well as helps in enrichment of cancer stem cell populations [6]. Aurora Kinase A inhibitor alisertib suppressed its oncogenic functions and was partly effective in clinical trials. However, it showed only limited efficacy in TNBCs [7], thus demanding the necessity to explore the underlying mechanisms of such insensitivity.

A recent report demonstrated an interesting observation related to the significant role played by polyploid giant cancer cells (PGCCs) during acquisition of resistance to PARPi inhibitor olaparib in ovarian cancer models [8]. PGCCs are known to be associated with high grade tumours and imparts resistance to chemotherapeutic success [9]. They form primarily by endocycling, nuclear fusion, or cytokinetic failure thus resulting in elevated genomic content and subsequent genomic instability [10]. This holds particular relevance in relation to Aurora Kinase A inhibition, considering the fact that Aurora Kinase A acts as a major gatekeeper protein in the process of spindle assembly, chromosomal segregation as well as cytokinesis, thereby influencing genomic integrity. Therefore, it is obligatory to look for the possible link between PGCC formation and alisertib sensitivity.

Reduced responsiveness towards Aurora Kinase A-specific small molecule inhibitor alisertib in TNBCs warrants the need to look for additional drugs that may offer advantage in improving its efficacy. We considered SHR antagonists for this purpose based on their unparalleled significance in deciding treatment strategy in breast cancers. TNBCs, although lack the detectable expression of classical SHRs yet they do express another important SHR, the Glucocorticoid receptor (GR); beyond the consideration of classical clinical practice. GR antagonized chemotherapy-induced cell death through enrichment

of Cancer Stem Cells (CSCs) and survival signalling cascades in TNBCs and associates with poor prognosis [11, 12]. Interestingly, GR antagonist mifepristone is also in clinical trial in TNBCs [13]. Mifepristone is further demonstrated in a recent study to effectively deplete PGCCs during acquired olaparib resistance in ovarian cancer [8].

Based on these previous observations, we felt intrigued to look for the effect of alisertib on the formation of PGCCs and its pursuing effect in determining alisertib sensitivity. We furthermore aimed to curb such insensitivity by use of GR antagonist mifepristone that may supplement the efficacy of alisertib in TNBCs.

2 Materials and methods

2.1 Reagents, chemicals and antibodies

The source/make of reagents/fine chemicals/antibodies required for this study were enlisted in Supplementary Table 1 (A & B).

2.2 Cell line maintenance

TNBC cell line MDA-MB-231 was cultured in Minimum Essential Media (MEM) supplemented with 10% FBS, along with added antibiotics (gentamycin 40 µg/ml, penicillin 100 units, and streptomycin 10 µg/ml). Cells were maintained in a humidified incubator containing 5% CO₂. MDA-MB-468 was maintained in Dulbecco's Modified Eagle Medium (DMEM, high glucose) medium supplemented with 10% FBS, along with added antibiotics (gentamycin 40 µg/ml, penicillin 100 units, and streptomycin 10 µg/ml).

2.3 Collection of patient samples

Triple negative breast cancer patients ($n = 15$), within the age group of 31–41 yrs were included in this study. Surgically resected breast tumour tissue and corresponding adjacent normal samples as identified by histopathologists were collected in accordance with the ethical guidelines. Ethical approval was taken from the Institutional Human Ethics Committee of Chittaranjan National Cancer Institute, Kolkata, India (Approval No# CNCI-IEC-SM-20, dt 28.7.2022). Informed Consent Form was taken from all the patients in accordance with the Helsinki Declaration.

2.4 Screening of patient records

To look for most commonly administered chemotherapeutics in TNBC patients, available records in hospital databases were checked and screened. All procedures were done with ethical permission and patient's identity was kept confidential.

2.5 Histoculture

Tumour tissue was washed in sterile PBS and thereafter cultured on gelatin sponges (Healthium Medtech private limited) presoaked with RPMI 1640 (HIMEDIA) media containing either vehicle (0.001% DMSO) or 100 nM alisertib for 48 h.

2.6 Alisertib treatment

Alisertib (Selleckchem) stock solution was prepared in DMSO. Cells were treated either with a sublethal dose of alisertib (100 nM) or with the corresponding vehicle (0.001% DMSO) for 48 h. For dose-dependent studies, 100 nM, 500 nM, and 1 µM doses were selected. For the pulse-holiday experimental setup, after insult of cells with each pulse of

100 nM, a recovery period of 20 days was maintained followed by another pulse of 100 nM alisertib. This was repeated for four consecutive pulses of alisertib insult. Finally, a pulse of 50 nM was administered to isolate the cell line featuring reduced sensitivity to alisertib.

2.7 Mifepristone treatment

Mifepristone (Sigma) stock solution was prepared in ethanol. Treatment was performed with a sublethal dose of 40 μ M mifepristone for 48 h. Corresponding vehicle used was ethanol (0.001%). For combination treatments, alisertib treatment was followed by a similar dose of mifepristone.

2.8 Morphological observation of cells

Morphological observations were performed under a phase contrast microscope (Nikon TMS) to identify and characterize PGCCs upon alisertib treatment. Mean percentage of polyploids in treatment vs. control were calculated by manual observation of giant cells in a field relative to the total no. of cells in that field and represented as mean $\pm$ S.D.

2.9 Flow cytometric analysis of DNA content by propidium iodide (PI) staining

Briefly, vehicle or alisertib-treated cells were harvested by trypsinisation, washed with ice-cold Phosphate Buffered Saline (PBS) and fixed by addition of dropwise chilled ethanol with mild vortexing. Following incubation on ice for an hour, ethanol was removed and washed twice with cold (PBS). Thereafter, cells were treated with RNase A (SIGMA-ALDRICH) and finally stained with PI (SIGMA) in dark for 15 min. Flow cytometric acquisition was done in a BD LSRFortessa flow cytometer. Analysis was performed with BD FACSDiva software. Polyploids were gated on the basis of DNA content by use of PE-A and FSC-H parameters.

2.10 BrdU incorporation assay

Cells were allowed to revive for 24 h post removal of alisertib. BrdU at a final conc. of 10 μ M was added to the media and left to incubate for 2 h for incorporation. Thereafter, media was removed and after washing, cells were fixed in cold ethanol. Following DNase (20 μ g) treatment, anti BrdU antibody was added and left to incubate overnight at 4 °C. Subsequently, FITC conjugated secondary antibody was added. Upon completion of incubation and subsequent washes, PI staining was done as mentioned before. Flow cytometric acquisition was done in a BD LSRFortessa flow cytometer. Analysis was performed with BD FACSDiva software. Polyploids were gated on the basis of DNA content by use of PE-A and FSC-H parameters.

2.11 TUNEL assay

Cells were treated with alisertib as mentioned earlier. Briefly, cells were fixed and permeabilised with permeabilisation buffer. Thereafter, cells were labelled with TDT and labelling mixture following manufacturer's protocol as described in In Situ Apoptosis Detection Kit (MK 500, TakaRa).

2.12 Single cell MTT assay

Cells were treated with alisertib as mentioned earlier. Following 5 days of revival 2×10^4 cells were seeded into 35 mm culture plates. After 24 h cells were incubated with MTT solution for 2 h in accordance to a previously published protocol [14]. Thereafter, microscopic observations were done under a (Nikon TMS) microscope to look for formazan crystal formation in PGCCs.

2.13 Immunofluorescence assay

Cells were seeded onto poly-L-lysine (Sigma) coated coverslips and thereafter treated as mentioned. Cells fixed with ice-cold acetone were stained with DAPI (Cell signaling technology) at a final concentration of 1 $\mu\text{g}/\text{ml}$ for 3 min. Mean percentage of polyploids in treatment vs. control was calculated by manual observation of giant cells in a field relative to the total no. of cells in that field and represented as mean $\pm$ S.D. For immunophenotyping, following fixation cells were blocked in Bovine Serum Albumin (BSA) and stained with respective primary antibodies overnight at 4 °C. Subsequently cells were washed in PBS and were incubated with corresponding FITC/PE-tagged secondary antibodies before staining with DAPI as mentioned above. Cells were observed under FV 4000 Olympus confocal microscope and images were acquired with Olympus cellSens software.

Histoculture tissues were fixed in 10% Neutral Buffered Formalin (NBF) and block preparation was performed according to standard procedures of dehydration and paraffin embedding. Blocks were processed into 5 μm sections in a rotary microtome, stretched and affixed on poly-L-lysine coated slides. Sections were deparaffinised in xylene and passed through descending grades of alcohol. Antigen retrieval was performed by heating in sodium citrate buffer (pH 6). Slides were washed in PBS and blocked in BSA. Thereafter, antibody staining was performed by incubation at 4 °C overnight. Subsequently, FITC/PE conjugated secondary antibodies were added and incubated at room temperature. DAPI staining was performed and sections were dehydrated in descending alcohol grades and xylene. Sections were mounted and observed under a FV 4000 Olympus confocal microscope, and images were acquired with Olympus cellSens software.

2.14 Limiting dilution and single cell cloning

Cells were treated and allowed to recover for 7 days. Subsequently, cells were trypsinized and cell number was determined in a hemocytometer. Cell suspension was serially diluted to a final conc. of 1 cell/100 μl which was added to the individual wells of a 96 well plate. Following 10 days of incubation, PGCCs were screened for formation of progeny under a phase contrast microscope (Nikon TMS).

2.15 Western blot analysis

Tumour as well as adjacent normal samples were washed in ice-cold PBS and thereafter lysed with RIPA lysis buffer in a mechanical homogenizer (REMI). Protein estimation was performed with Bradford reagent (HIMEDIA). An equal amount of protein was loaded into 10% SDS-Polyacrylamide gels for electrophoretic separation. Thereafter, proteins were blotted onto Polyvinylidene fluoride (PVDF) membrane in a semi dry transfer apparatus (Bio-Rad). Blocking was done with 5% BSA in Tris buffered saline

with tween 20 (TBS-T). Membranes were probed with anti AURKA mouse monoclonal antibody/anti β -actin mouse monoclonal antibody overnight at 4 °C. After subsequent washes with TBS-T anti mouse alkaline phosphatase conjugated secondary antibodies were added and incubated for 2 h. After subsequent washes, bands were visualised by addition of BCIP-NBT solution (SIGMA-ALDRICH).

2.16 Mammosphere forming assay

Equal number of ali pulsed (5) cells either treated with either alisertib (100 nM) or mifepristone (40 μ m) or left untreated were seeded in ultralow attachment plates (Corning) in DMEM-F12 medium (Himedia) supplemented with hEGF (20 ng/ml), bFGF (10 ng/ml) and B27 (1x) and maintained for 7 days. Formed spheres were imaged under a microscope (Olympus CKX 41). Images were acquired with infinitycapture software. Mammosphere forming efficiency was calculated as the (no. of mammospheres formed / no. of cells seeded) x 100%.

2.17 MTT assay

Equal no. of cells of control or ali pulsed (5) cells were seeded in wells of a 96-well plate and treated with either vehicle or 25nM, 50nM, and 100 nM of alisertib, respectively for 48 h. Thereafter, MTT (SRL) solution was added to each well and incubated according to standard procedures. Formed formazan crystals were dissolved in DMSO and absorbance was recorded at 570 nm in a Tecan microplate reader infinite M200. Absorbance values were interpreted in terms of the percentage of viable cells relative to a control.

2.18 Colony forming assay

Briefly MDA-MB-231 cells were treated with alisertib/ alisertib-mifepristone as mentioned above. Equal number of cells were thereafter seeded and allowed to grow for 10 days to form colonies. Formed colonies were fixed with methanol: acetic acid (3:1) and stained with 0.1% crystal violet for subsequent visualization.

2.19 Histopathologic examination by H&E staining

Surgically resected tumour/histoculture tissues were fixed in 10% Neutral Buffered Formalin (NBF) and processed as mentioned earlier. Sections were deparaffinised in xylene and passed through descending grades of alcohol and finally stained with hematoxylin and eosin. Following dehydration in ascending grades of alcohol and xylene, mounting was done with Dibutylphthalate Polystyrene Xylene (DPX), and sections were observed under a microscope (Zeiss primostar). PGCCs in stained histoculture sections were counted relative to the number of cells in a field and expressed as percentage values.

2.20 In silico Pharmacokinetic analysis and toxicity prediction

Percent similarity with Lipinski's rule of five and bioavailability scores were calculated and compared between alisertib and commonly administered chemotherapies in Target-Net [15]. Bioavailability radar plots were acquired for alisertib and commonly used chemotherapeutics from swissADME [16]. Toxicity prediction for selected parameters of organ toxicity (hepatotoxicity, nephrotoxicity, neurotoxicity, cardiotoxicity, respiratory toxicity) were performed in proTox 3.0 [17] and represented as network charts and toxic radar charts.

2.21 Pan cancer expressional profiling in GEPIA

Top nine kinases with the highest CKI scores were assessed in GEPIA [18] for differential expression status in all mentioned cancers in TCGA.

2.22 Comparative expressional analysis

AURKA expression (log2 standardised mRNA level) was compared between healthy (92 patients), tumour adjacent (89 patients) and tumour tissues (743 patients) and as well as between TNBC (578 patients) and non-TNBCs (87 patients) in bc-GenExMiner [19].

2.23 Clinical kinase indexing

Clinical kinase index (CKI) scores [20] of top kinases in the target developmental level Tchem were acquired to identify clinically relevant kinases in breast cancer with the potential of acceptance in clinical use in near future.

2.24 Druggability scoring

Druggability probability of AURKA was determined in SPIDER [21]. Drug gene interaction scores for AURKA were calculated in DGIdb [22]. Drugs with interaction scores greater than 0.1 were further checked for their clinical trial status.

2.25 Differential expression analysis in publicly available datasets

Differential expression analysis in two independent public datasets GSE21422 and GSE61304 available from NCBI Gene expression omnibus (GEO) database (<https://www.ncbi.nlm.nih.gov/geo/>) were performed in Geo2R [23] Common differentially upregulated genes were identified in venny 2.0 and represented as a venn diagram.

2.26 Survival analysis

Impact of kinases that are differentially overexpressed in more than 50% of TCGA cancers on overall survival (OS) were assessed in KM plotter. Following datasets PTK6 (1553114_a_at), PLK1 (202240_at), PGK1 (227068_at), EPHB2 (209588_at), AKT1 (207163_s_at), MAPK12 (1556341_s_at), CDC2(203213_at), CT84(219148_at), AURKA (208079_s_at) were assessed for OS. Impact of AURKA on Distant Metastasis Free Survival (DMFS), Post progression survival (PPS), Relapse Free Survival (RFS) were also assessed using dataset (208079_s_at).

2.27 Statistical analysis

All statistical analysis were performed in GraphPad prism 8.0.1. All experiments were performed in replicates. Data are represented as Mean $\pm$ S.D. For datasets consisting of two groups student's t test was performed, while for more than two groups, one-way anova was performed. $P < 0.05$ was considered statistically significant.

3 Results

3.1 Aurora Kinase A (AURKA) is a major candidate of the prognostically relevant targetable Kinome in breast cancer

Breast cancer (BC) kinome was organized initially based on clinical kinase index (CKI) scoring, with an endeavor to search for those kinases that have potential clinical value. The result as depicted in Fig. 1A showed that Aurora Kinase A or AURKA ranked as one

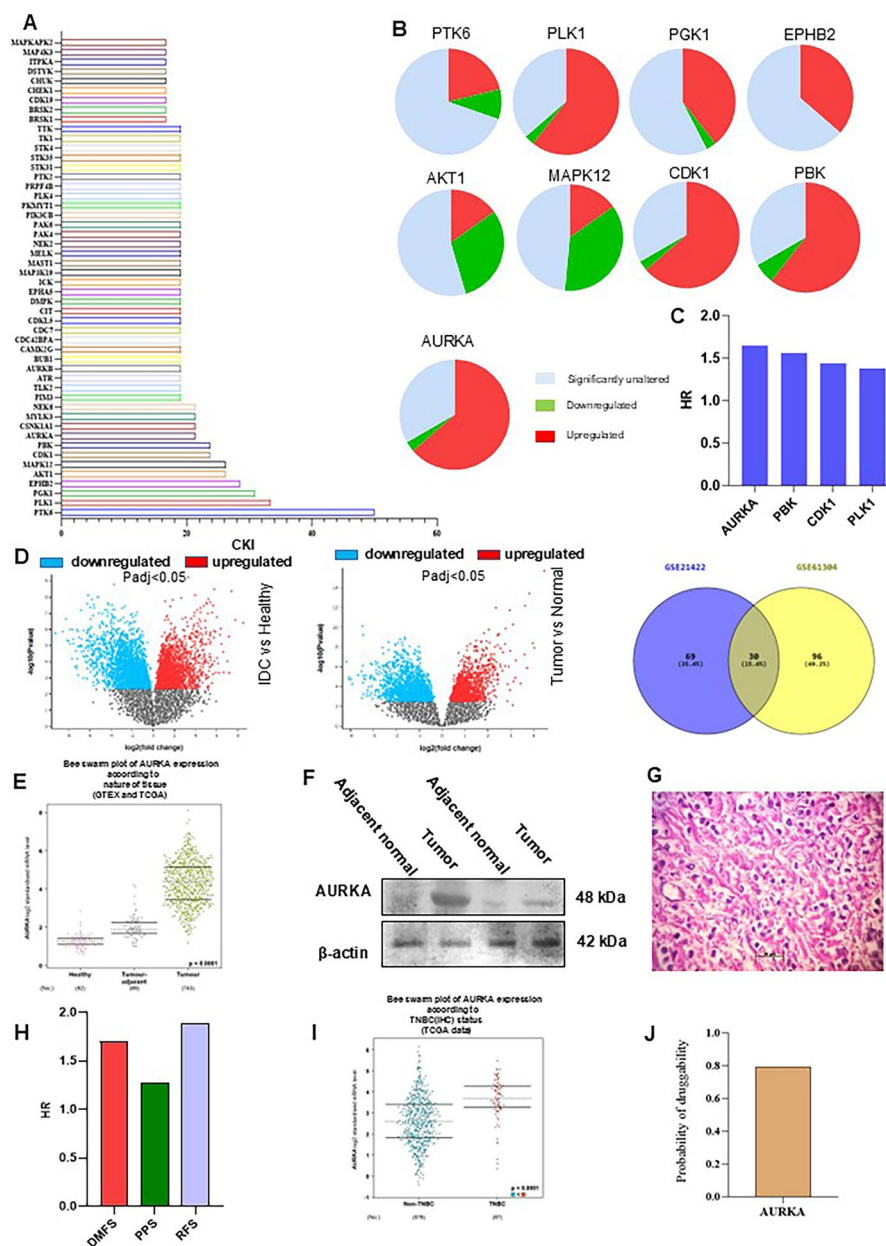


Fig. 1 Data mining of prognostically relevant targetable kinome in breast cancer flagged Aurora kinase A (AURKA) as a major candidate. **A** Graphical hierarchical representation on the basis of CKI scores of clinically relevant kinome. **B** Pie-chart depicting the expressional status of top kinases with clinical relevance in TCGA-defined cancer types. **C** Bar graph denoting the impact of selected kinases, which are overexpressed in $\geq 50\%$ TCGA cancers on overall survival. The data has been represented in terms of Hazard Ratio. **D** Volcano plots (Left panel) illustrating differentially expressed genes between normal and tumour breast as obtained from two different datasets. Red indicates the set of upregulated genes, while the blues are the downregulated genes. Venn diagram (Right panel) showing the common upregulated genes between the two datasets. **E** Bee swarm plot of differential AURKA expression in breast tissues (normal vs. adjacent normal vs. tumour). **F** Representative Immunoblots demonstrating differential expression of AURKA in paired tumour and adjacent normal tissues. **G** Histological characterization of breast tumour by H&E staining. Original magnification of the objective: 40X. **H** Impact of AURKA on DMFS, RFS and PPS as evident from the obtained scores of Hazard Ratio. **I** Bee swarm plot of AURKA expression in non-TNBC vs. TNBCs. **J** Bar graph displaying the likelihood of druggability of AURKA

of the top kinases with a CKI value of 21.43; indicating its clinical importance. Based on this CKI scoring, top nine kinases were selected/chosen for further assessment of their expressional status. This was attempted with the notion that kinases that are overexpressed in majority of the cancers might have a universal implication in malignant progression. Pan-cancer comparative expressional profiling revealed PLK1, CDK1, PBK and AURKA as the most relevant kinases based on their overexpression in more than 50% of the TCGA cancer types (Fig. 1B). These kinases were further scanned for their impact on overall survival of breast cancer patients. Kaplan-Meier plot (Supplementary Fig. 1A–D) clearly showed that with a hazard ratio (HR) of 1.65, Aurora Kinase A had the highest impact on overall survival over other mentioned kinases (Fig. 1C), affirming its prognostic significance. Next, we felt curious to understand whether Aurora Kinase A was indeed a commonly upregulated gene in breast cancer. For that purpose, two independent datasets (GSE21422 and GSE61304) were analyzed in GEO2R, where differential expression of Aurora Kinase A between the normal breast tissue and malignant counterpart was evident. Venn diagram representing the common thirty-upregulated genes from both datasets elucidated Aurora Kinase A as a commonly overexpressed gene in breast cancer (Fig. 1D).

To further validate and assess the alteration in the expression of Aurora Kinase A in breast cancer, its expressional profiles were compared among normal, adjacent normal and cancerous breast tissues using TCGA and GTEX databases. Mined results demonstrated that in normal breast tissue Aurora Kinase A expression was low which was greater in adjacent normal while substantially elevated in cancerous breast tissues (Fig. 1E). We further re-assured Aurora Kinase A status in cancer tissues versus paired adjacent normal tissues by western blotting (Fig. 1F). Histological observations of the cancer tissues obtained were primarily of invasive ductal carcinoma type (Fig. 1G) with typical features characterized by loss of normal tissue architecture and presence of hyper-chromatic nucleus. The brief details of study subjects were described in Supplementary Table 2 (A & B). We next focused on checking the effect of low / high Aurora Kinase A on distant metastasis free survival (DMFS), post-progression survival (PPS), and relapse free survival (RFS) using available datasets of Kaplan-Meier plotter or KM plotter. Hazard Ratio (HR) scores as obtained were plotted graphically (Fig. 1H) which demonstrated the negative impact of Aurora Kinase A elevation on DMFS, PPS, RFS with HRs of 1.89, 1.7, and 1.27, respectively. The original KM plots were provided in Supplementary Fig. 1E. Next, it was felt necessary to look for alterations in Aurora Kinase A status among different BC subtypes. Data mining results displayed significant augmentation of Aurora Kinase A in the most aggressive TNBC subtypes (Fig. 1I). These results clearly justified the relevance of this particular kinase in the context of breast cancer prognosis. We hence estimated the druggability of Aurora Kinase A, wherein it's high probability as a drug target was apparent (Fig. 1J). These results thus uphold the need for an exploration of the prospective Aurora Kinase A targeting molecules.

3.2 Alisertib being an Aurora Kinase A targeting agent showed great clinical importance in breast cancer

Concerning the significance of Aurora Kinase A as a druggable target in breast cancer, we screened for available drugs that target Aurora Kinase A using DGIdb webserver, available for predicting drug-gene interactions. Calculated drug-gene interaction scores

were tabulated in Supplementary Table 3. On the basis of these scores, the drugs that surpassed the cut-off of 0.1 with higher specificity for Aurora Kinase A, were further assessed for their success in clinical trials. Among the series of Aurora Kinase A Inhibitors (AKIs) developed over the decades, alisertib was the only targeting agent to reach up to the phase III clinical trials (Supplementary Table 4).

We therefore aimed to assess the drug-likeness properties and bioavailability of alisertib compared to other routinely used chemotherapeutics. To achieve this, we started with searching for treatment regimen followed in our institute and primarily found doxorubicin, docetaxel, epirubicin, cyclophosphamide, 5-fluorouracil and paclitaxel as the routine choice of chemotherapeutics. The drugs were listed in Fig. 2A.

Alisertib was thereafter compared with these drugs for percent match with Lipinski's rule of five, a thumb rule that helped to predict the efficacy of the drugs using TargetNet web server. Interestingly, it was observed that alisertib performed better than most of the drugs barring cyclophosphamide and 5-fluorouracil (Fig. 2B). Bioavailability radar plots further depicted satisfactory bioavailability of alisertib; obeying most of the bioavailability parameters (Fig. 2C). Furthermore, alisertib performed as good as or better than the routine chemotherapeutic drugs as demonstrated from its high bioavailability a score of 0.56 (Fig. 2D). This result corroborated its good or even better performance compared to other routinely used chemotherapeutics.

Next, we evaluated the toxicological properties of alisertib in comparison to the other above-mentioned drugs using an *in silico* approach. Alisertib showed no cardiotoxicity (Inactive cluster probability > 0.7), but exhibited low levels of nephrotoxicity, diliohepatotoxicity and neurotoxicity (Active cluster probability < 0.7). However, notable respiratory toxicity (Active cluster probability > 0.7) was predicted (Fig. 2E). Docetaxel, paclitaxel, doxorubicin, epirubicin on the other hand showed considerable nephrotoxicity, neurotoxicity and respiratory toxicity (Active cluster probability > 0.7) while cyclophosphamide and 5-Fluorouracil were predicted to show high probability (Active cluster probability > 0.7) of being neurotoxin as well as respiratory toxins (Supplementary Fig. 2). Corresponding toxicity radar charts illustrating organ toxicity profiles of selected molecules in comparison to other active molecules/class were depicted in (Supplementary Figs. 3–6). Alisertib therefore, seems to hold prospect as an agent to target AURKA in breast cancer.

3.3 Alisertib being a selective Aurora Kinase A inhibitor brought about PGCC enrichment that fecundate daughter cells

As Alisertib was found to be a promising agent with high bioavailability score and tolerable toxicity profile, we felt curious to examine the impact of the same on triple negative MDA-MB-231 cells. With that intention, MDA-MB-231 was challenged with alisertib dose of 100 nM as described in the Methodology Section.

Surprisingly, specific kind of giant cells with atypical morphology were clearly visible upon treatment (Fig. 3A, left panel). Giant cells versus total number of cells in each field were calculated under the phase contrast microscope and accordingly percentage of giant cell enrichment was plotted graphically expressed as mean $\pm$ S.D. (Fig. 3A right panel).

Microscopic observation of such specialized cells were found to be morphologically heterogeneous with either a single large irregular-shaped nucleus with flattened

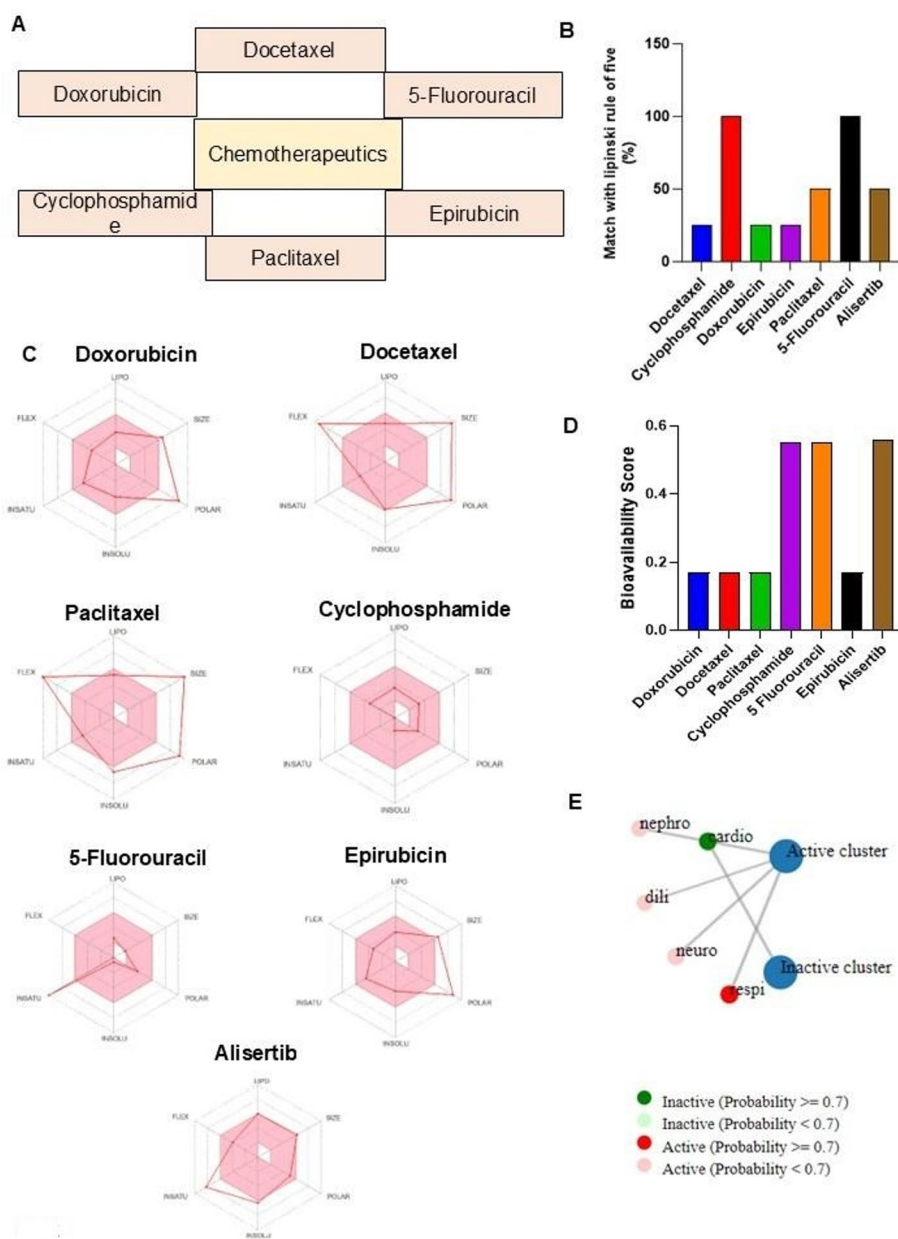


Fig. 2 Clinical implication of Alisertib in breast cancer. **A** Showing the routinely used drugs as screened from patient databases. **B** Comparative drug likeliness analyzed on the basis of Lipinski's rule of five among alisertib and other routinely used chemotherapeutics. **C** Bioradar plot indicating the bioavailability parameters of alisertib along with other routine chemotherapeutics. **D** Calculated bioavailability scores acquired from Swissadme have been plotted graphically. **E** Network chart displaying the predicted organ toxicities (hepatotoxicity, nephrotoxicity, neurotoxicity, cardiotoxicity, respiratory toxicity) of alisertib as carried out in ProTox 3.0

appearance, or with multiple small nuclei, while others depicted a neuron like morphology with cytosolic extensions (Fig. 3B). DAPI stained imaging of such cells further ensured the presence of multinucleated giant cells following alisertib insult (Fig. 3C). Such features are nothing but typical characteristics of polyploid giant cancer cells (PGCCs). To further ascertain that such population is indeed polyploid in terms of genomic content, flow cytometric analysis was performed. The result obtained definitively showed a doubling of the polyploid population post-alisertib treatment. Such

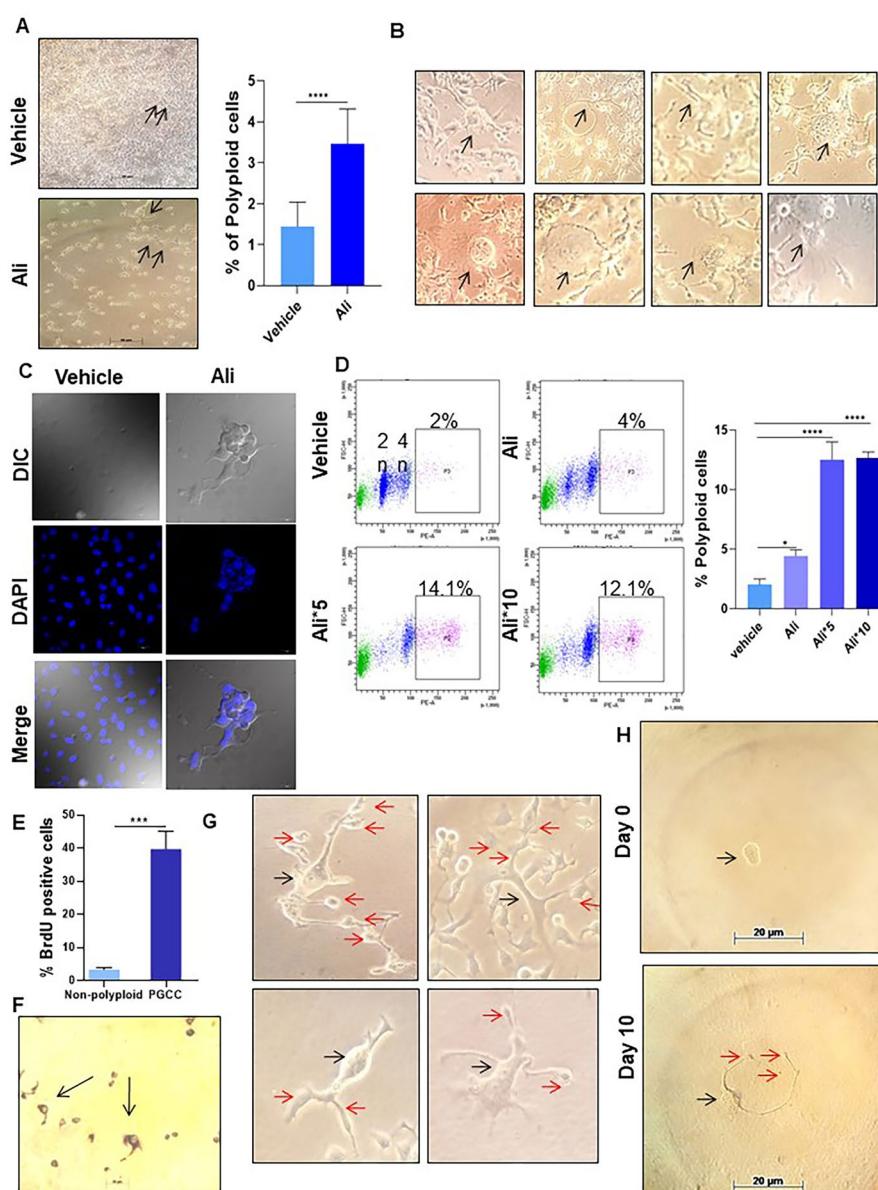


Fig. 3 Polyploid Giant Cancer Cell (PGCC) enrichment evoked by alisetib was succeeded by the formation of PGCC progenies in MDA-MB-231. **A** Microscopic observation (Right panel) of morphological changes in alisetib (100 nM) treated MDA-MB-231 cells compared to untreated counterpart. Scale Bar: 50 μ m, original magnification of the objective: 20X. Total number of polyloid population was counted in throughout the field and (% polyloid population) has been represented graphically. The experiment was conducted in triplicates and mean percentage value has been plotted. $p < 0.0001$ denoted significant increase in polyploidy population compared to the vehicle control. **B** Zoomed images (Not in scale) depicting atypical morphological characteristics of polyloid giant cells under a phase contrast microscope. **C** DAPI stained confocal microscopic images showing multinucleated PGCCs in alisetib-treated cells. Scale Bar: 20 μ m, original magnification of the objective: 60X. **D** Dot plot representation (Left panel) of the flow cytometric results showing dose-dependent effect of alisetib on PGCC enrichment. (Ali: 100 nM, Ali*5: 500 nM, Ali*10: 1 μ M). Results have been represented graphically in terms of % polyloid cells (Right panel). **E** Column graph showing the percentage of BrdU positive non-polypliods and polypliods upon alisetib (100 nM) treatment. **F** Representative microscopic image depicting formazan crystal formation by PGCCs. Original magnification of the objective: 40X, Scale Bar: 20 μ m. **G** Phase contrast microscopic zoomed images (not in scale) distinctly showing daughter cell production. Original images have been inserted in the supplementary Fig. 7. Black arrow indicates mother cells, while red arrow denotes progeny. **H** Phase contrast microscopic images showing the ability of a single PGCC to form daughter cells. Black arrow indicates mother cells, while red arrow denotes progeny. Original magnification of the objective: 40X, Scale Bar: 20 μ m

effect was clearly dose dependent and thus validated the enrichment of PGCCs as a direct consequence of alisertib administration (Fig. 3D) wherein an almost incremental increase in the PGCC percentage was found with the dose administered. However, we did find a slight decrease in the PGCC percentage at the highest dose administered that might be presumed to be a result of off-target effects at the highest dose administered. Enrichment of such cells further demanded to explore whether these cells eventually die or possess the capacity to replicate. BrdU positivity of such polyploid cells revealed that they maintain replicative ability even after alisertib treatment. PGCCs further showed a higher percentage of BrdU positivity compared to non-polyploids as represented graphically (Fig. 3E). Flow cytometric dot plots have been incorporated in Supplementary Fig. 7A. To further study the long-term fate of these PGCCs, we performed single cell MTT assay to determine their viability and metabolic activity. The result (Fig. 3F) clearly depicted maintenance of metabolic viability of some PGCCs during the recovery period after alisertib withdrawal (Fig. 3F). Some PGCCs however, maintained Ki-67 negativity, probably indicative of their quiescence in face of alisertib challenge. TUNEL positivity was also noted at least for some of the PGCCs demonstrating that some PGCCs may also undergo cell death (Supplementary Fig. 7B and C). These findings further instigated to find out whether the surviving PGCCs could have also reached a dead end of their life cycle or they retain their ability to undergo cell division and produce progenies. Cells were tracked after withdrawal of alisertib and following 10 days of alisertib removal, we observed the generation of daughter cells by some of these PGCCs which underwent atypical cell divisions where daughter cells originated from branches of the mother cells or by budding (Fig. 3G). Corresponding captured fields under phase contrast microscope were shown in Supplementary Fig. 8. We next isolated a single PGCC by means of serial dilution and seeded it in a 96 well plate. The plate was tracked for daughter cell formation, which showed that some PGCCs showed single cell clonogenicity as marked by formation of daughter cells (Fig. 3H). This result signified that inspite of low abundance of PGCCs as obtained in our study, they may be potent enough to lay the foundation of tumor cell colonies. This raised the possibility that PGCCs may be an escape mechanism adopted by cancer cells to overcome periods of alisertib-stress that is characterised by subsequent recolonization by PGCC derived daughter cells.

3.4 Enriched PGCCs escaping alisertib challenge retained the expression of markers associated with EMT and stemness features

To assess the ability of PGCCs to cope with alisertib stress, we aimed to look for if they retained the ability to express key proteins associated with stemness like features and EMT even in face of alisertib challenge. Remarkably, PGCCs expressed ALDH1A1, a key determinant of cancer stemness despite alisertib treatment. This typical molecular characteristics of PGCCs indeed indicated towards their stem-like potential (Fig. 4A). This observation was supported by augmented expression of Oct4 under the same condition which further reinstated the stemness potential of PGCCs (Fig. 4B). We further extended our study by checking the expression of the filamental protein vimentin in treated cells owing to its involvement in EMT process. Herein, the results demonstrated substantial expression of vimentin in the extensive filamental network of the PGCCs, raising the possibility of their integral participation during the event of EMT (Fig. 4C). Immunofluorescence images manifested that some of the PGCCs were potent enough

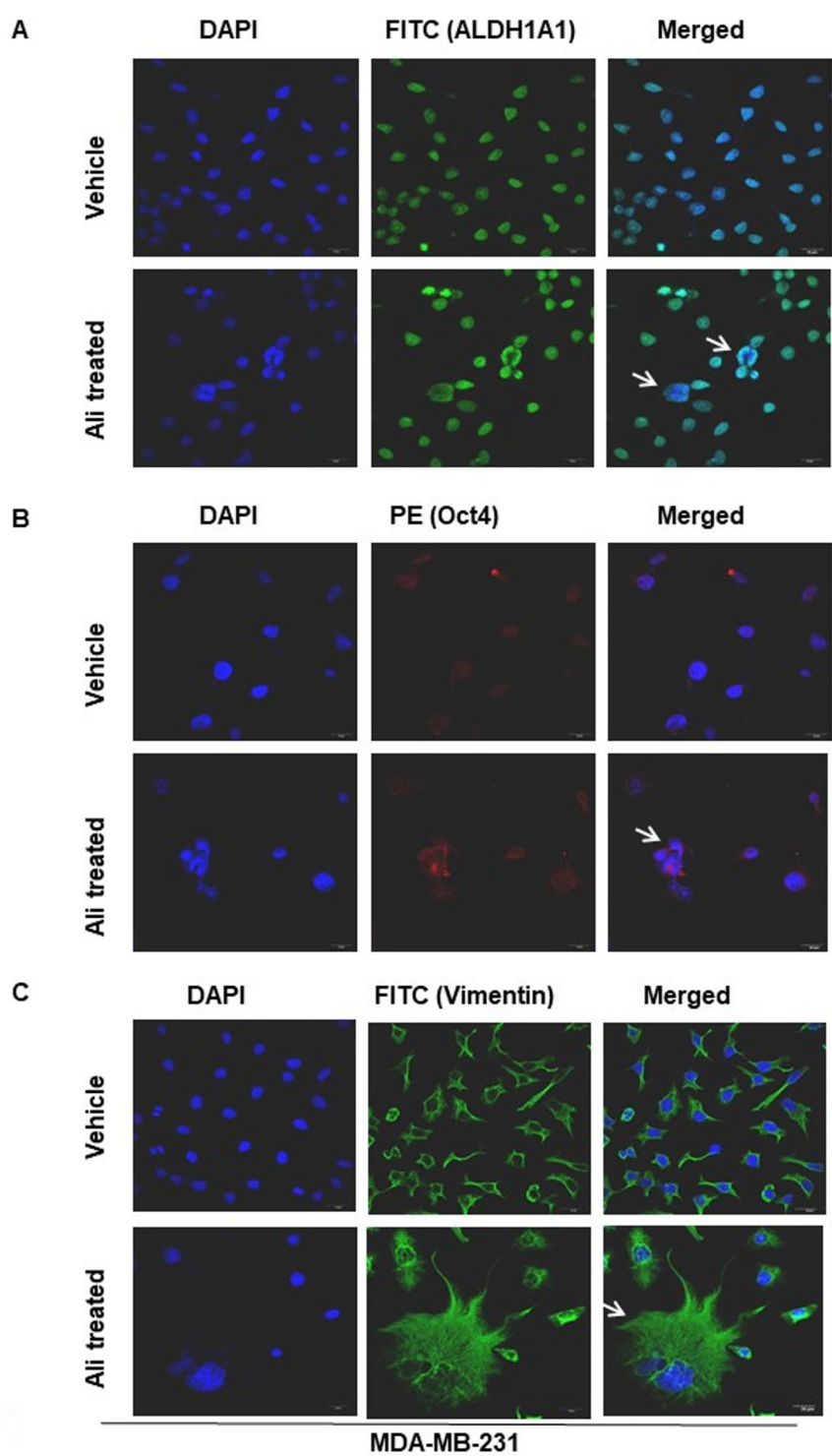


Fig. 4 Enrichment of EMT and stemness features in alisertib escaping PGCCs regardless of drug insult in MDA-MB-231. Immunofluorescence images of stemness markers **(A)** ALDH1A1, **(B)** Oct4 and **(C)** EMT marker vimentin. Original magnification: 60X, scale bar: 20 μ m

to generate daughter cells irrespective of alisertib stress. Additionally, the daughter cells formed from PGCCs also retained the expression of vimentin (Fig. 4C). These cumulative observations indicated the remarkable property of PGCCs and probably their progenies to sustain the expression of EMT and stemness markers even after challenging them with alisertib.

3.5 Functional Inhibition of Aurora Kinase A by alisertib resulted in increased abundance of PGCCs in MDA-MB-468

Alisertib induced PGCC enrichment in TNBC cell line MDA-MB-231 left us intrigued to examine the impact of the same on another triple negative cell line. For this purpose, MDA-MB-468 was also challenged with alisertib at a dosage of 100 nM as described in the Methodology Section.

Here also, atypical giant cells with diverse morphological attributes were observable upon treatment (Fig. 5A, left panel). Giant cells versus total number of cells in each field were calculated under the phase contrast microscope and accordingly, percentage of giant cell enrichment was plotted graphically and expressed as mean $\pm$ S.D. (Fig. 5A right panel).

Microscopic observation of such cells revealed morphological diversity with either a single large nucleus, or with multiple small nuclei, while others depicted a neuron like morphology (Fig. 5B). DAPI staining further confirmed the enhanced presence of multinucleated giant cells following alisertib treatment, as shown in Fig. 5C, and depicted graphically (Right panel) as the increase in percentage of polyploid cells. Flow cytometric analysis was further performed to validate the results. The results demonstrated a definitive increase in the percentage of polyploid cells post-alisertib treatment (Fig. 5D). To further study the long-term fate of these PGCCs, we performed single cell MTT assay to determine their viability. The result (Fig. 5E) depicted maintenance of metabolic viability of PGCCs during the recovery period after alisertib withdrawal as observable from formation of formazan by PGCCs. These observations further affirm the potential role of PGCCs as a common adaptive feature adopted by TNBC cells to endure alisertib stress.

3.6 EMT and stemness marker expression persisted in PGCCs following alisertib administration in MDA-MB-468 cells as well

To assess the capacity of PGCCs to withstand alisertib-induced stress, we investigated whether they retained the ability to express key proteins associated with stemness and epithelial–mesenchymal transition (EMT). Remarkably, PGCCs continued to express ALDH1A1, a well-established marker of cancer stemness, even following alisertib exposure (Fig. 6A). This molecular profile was suggestive of the persistence of stem-like characteristics in PGCCs even after alisertib administration. Supporting this observation, we also detected retained expression of Oct4, another stemness marker in these cells further reinforcing the stemness potential of PGCCs (Fig. 6B).

To explore their EMT-related features, we examined the expression of vimentin, a filamentous protein crucial to the EMT process. Immunofluorescence imaging revealed that at least some of the PGCCs show vimentin expression, suggesting their potential involvement in EMT-related mechanisms (Fig. 6C).

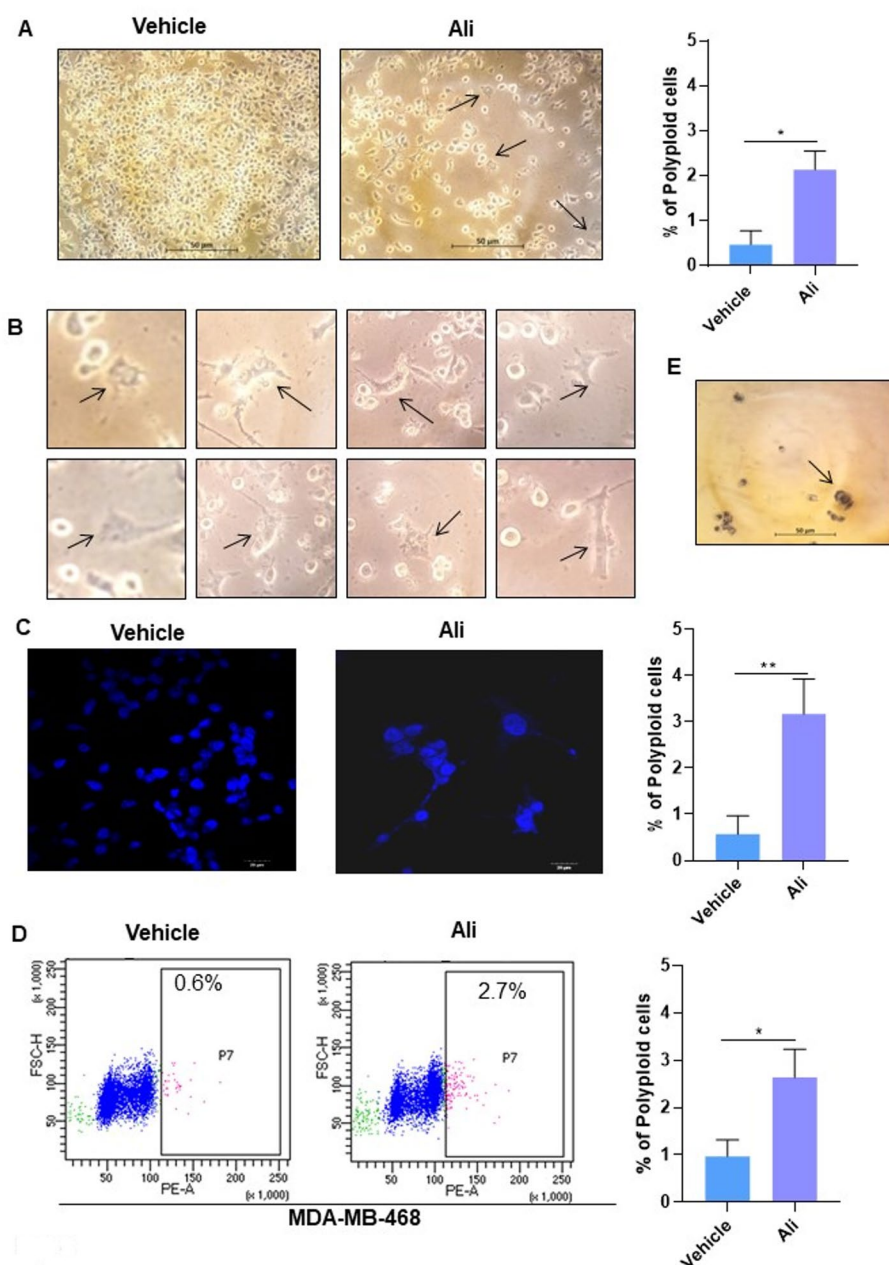


Fig. 5 Polyploid Giant Cancer Cell (PGCC) enrichment evoked by alisertib was succeeded by the formation of PGCC progenies in MDA-MB-468. **A** Microscopic observation (Right panel) of morphological changes in alisertib (100 nM) treated MDA-MB-231 cells compared to untreated counterpart. Scale Bar: 50 μ m, original magnification of the objective: 40X. Total number of polyploid population was counted in throughout the field, and (% polyploid population) has been represented graphically. The experiment was conducted in triplicates and mean percentage value has been plotted. $p < 0.05$ denoted significant increase in polyploidy population compared to the vehicle control. **B** Zoomed-in phase contrast images (not to scale) highlighting atypical morphological characteristics of PGCCs, including enlarged size and irregular nuclear morphology. **C** DAPI-stained confocal microscopy images showing increased abundance of PGCCs following alisertib treatment. Scale bar: 20 μ m; original magnification: 60X. PGCCs were quantified relative to total cell number, and the percentage is graphically represented as mean $\pm$ SD. Statistical significance was set at $p < 0.05$. **D** Dot plot representation (Left panel) of the flow cytometric results showing dose-dependent effect of alisertib on PGCC enrichment. (Ali: 100 nM). Results have been represented graphically in terms of % polyploid cells (Right panel). **E** Representative phase contrast image demonstrating viable PGCCs as indicated by formazan staining. Scale bar: 50 μ m; original magnification: 40X

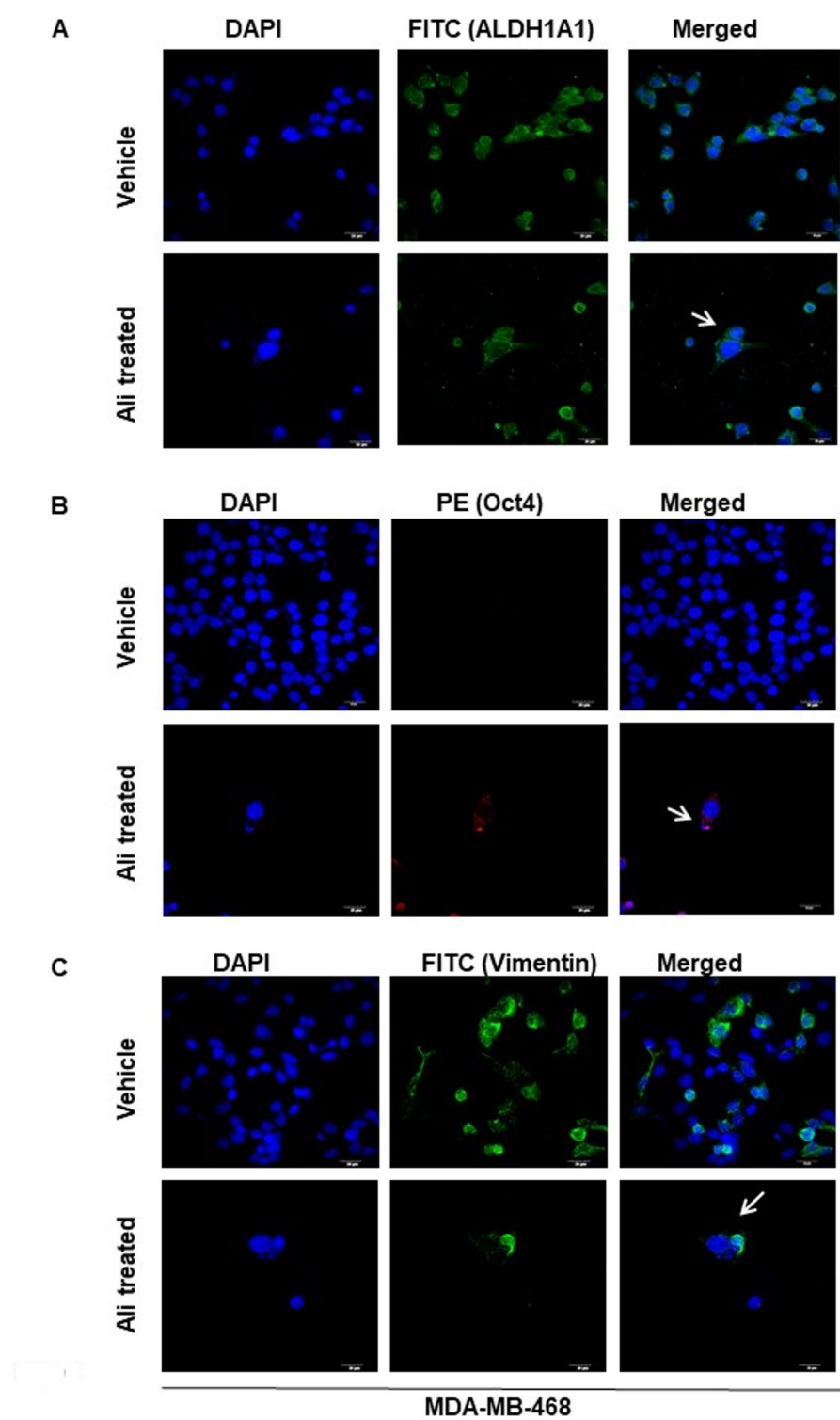


Fig. 6 EMT and stemness features are enriched in alisertib escaping PGCCs in MDA-MB-468. Confocal microscopic images of stemness markers **(A)** ALDH1A1, **B** Oct4 and **C** EMT marker vimentin. Original magnification: 60X, scale bar: 20 μ m

Taken together, these findings underscore the remarkable ability of PGCCs and potentially their progeny to maintain the expression of both stemness and EMT markers even in the face of alisertib-induced stress.

3.7 TNBC tissues reciprocated the in vitro observations in terms of PGCC enrichment subject to alisertib treatment

We next explored whether alisertib induced enrichment of PGCCs as observed in in vitro condition is at all reflected in the TNBC patient scenario. Patient tumour derived histocultures were therefore exposed to alisertib and checked for percentage of polyploid cells relative to untreated counterpart ($n = 3$). Brief outline of the histoculture procedure had been schematically illustrated (Fig. 7A). Percentage of PGCCs were calculated as the number of polyploids relative to total number of cells in each of the field of the H&E stained tissue sections and compared against the untreated control. We observed distinct PGCC enrichment due to alisertib insult even in the ex vivo scenario as evident from the representative histological images (Fig. 7B-left panel). Corresponding graphical representation of PGCC percentages were shown in Fig. 7B-right panel). Previous results displayed in Fig.[4,6] intrigued us to check for the expression of the same EMT and stemness markers in the histoculture model. Immunofluorescence images here also revealed that alisertib failed to abolish the expression of ALDH1A1, Oct4 and Vimentin (Fig. 7C) in the PGCCs of treated histocultures thus revalidating our in vitro findings.

3.8 PGCC-derived daughter cells evading the successive pulses of alisertib insult fostered the acquisition of compromised sensitivity towards alisertib

Our previous findings demonstrating that alisertib induced formation of PGCCs retained the ability to produce subsequent progenies prompted us to track the fate of such cells in a situation partially mimicking the clinical setup. A typical regimen of repeated drug pulsing intervened by drug holidays, that are often followed in clinical settings was partly replicated here in in vitro set up. The study design of pulsed alisertib administration has been depicted schematically in Fig. 8A.

Flow cytometric assay (Fig. 8B) was undertaken to assess the percentage of polyploid cells revealed that initially during the recovery phases following first, second or third pulses of alisertib exposure hardly any change in the polyploidy formation were noted. Nevertheless, when the cells were re-exposed for fourth pulse an appreciable increase in the polyploid population was evident. These results pointed towards the acquirement of an adaptive response by these cells, which would probably make these cells capable enough to confront alisertib challenge more efficiently.

We therefore exposed these cells to a comparatively lower dose of alisertib (50 nM) with a notion that the surviving clones will feature diminished alisertib sensitivity. To affirm that, we performed MTT assay by challenging both parental MDA-MB-231 cells and pulsed-(5) counterpart with incremental doses of alisertib. The absorbance values of developed formazan revealed that in parental cells alisertib performed better compared to the pulsed-(5) counterpart. This observation clearly confirmed that the cells exposed to repetitive alisertib challenge have ultimately developed decreased sensitivity towards the same (Fig. 8C).

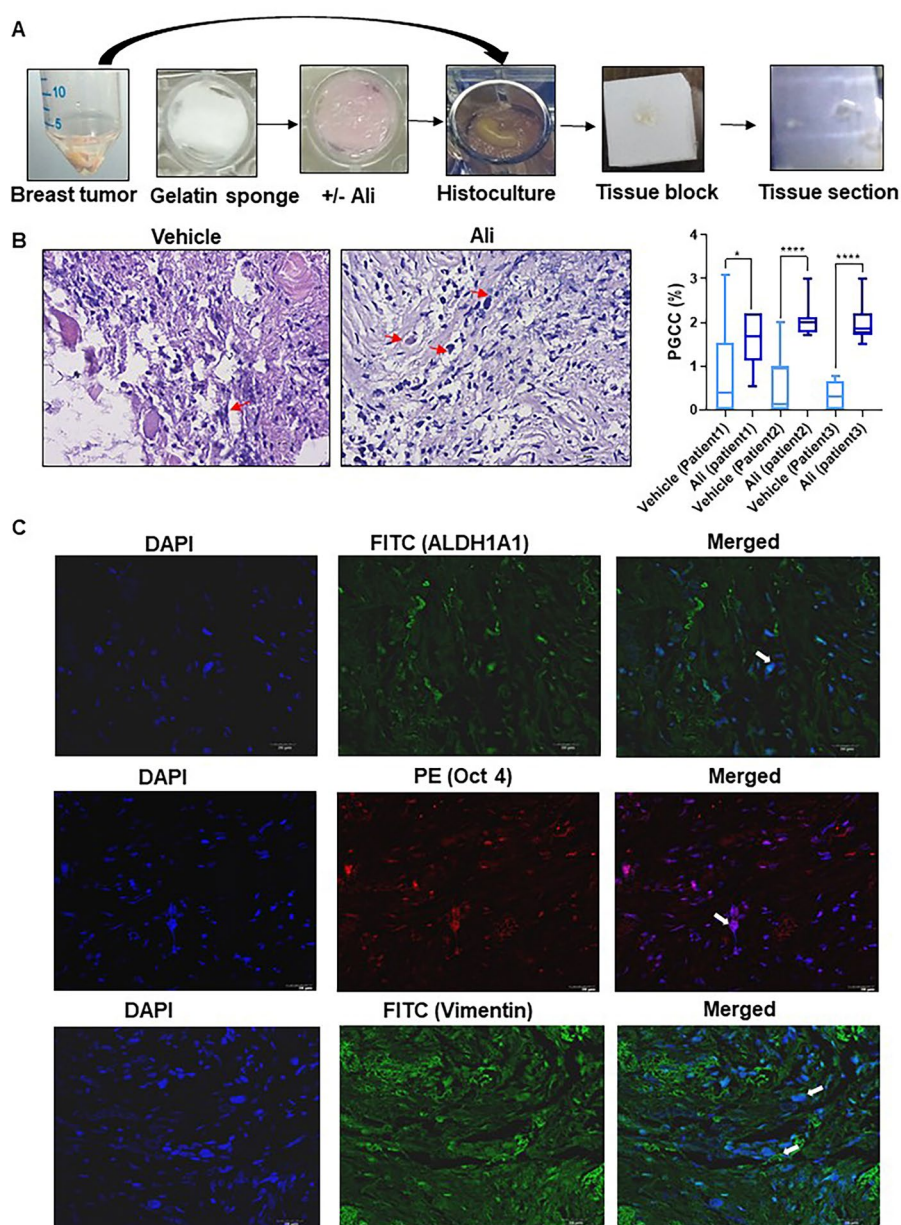


Fig. 7 Detection of PGCCs in TNBC-derived histocultures. **A** A schematic outline describing the histoculture process, starting from tumor tissue collection to microtome sectioning. **B** Representative images of H & E stained sections showing prevalence of PGCCs in histocultures treated with alisertib (left panel); Corresponding box plots denoting % of PGCCs in untreated vs. ali-treated condition (right panel). Original magnification: 40X, scale bar: 10 μ m. **C** Expression of ALDH1A1 (upper row), Oct4 (middle row) & Vimentin (bottom row) as evident from representative immunofluorescence images of treated histoculture sections. Original magnification: 60X, scale bar: 20 μ m

3.9 Mammosphere forming ability in TNBC cells left unaddressed by alisertib but not by mifepristone

Our previous results demonstrated reduced alisertib sensitivity in MDA-MB-231 cells upon repeated pulsing. We hence looked for other alternative strategies to overcome this insensitivity. Considering the fact that Glucocorticoid Receptor (GR) is intricately involved in maintenance of cancer stem cell population with drug resistant properties, we checked for GR status in alisertib treated MDA-MB-231 derived PGCCs as well as

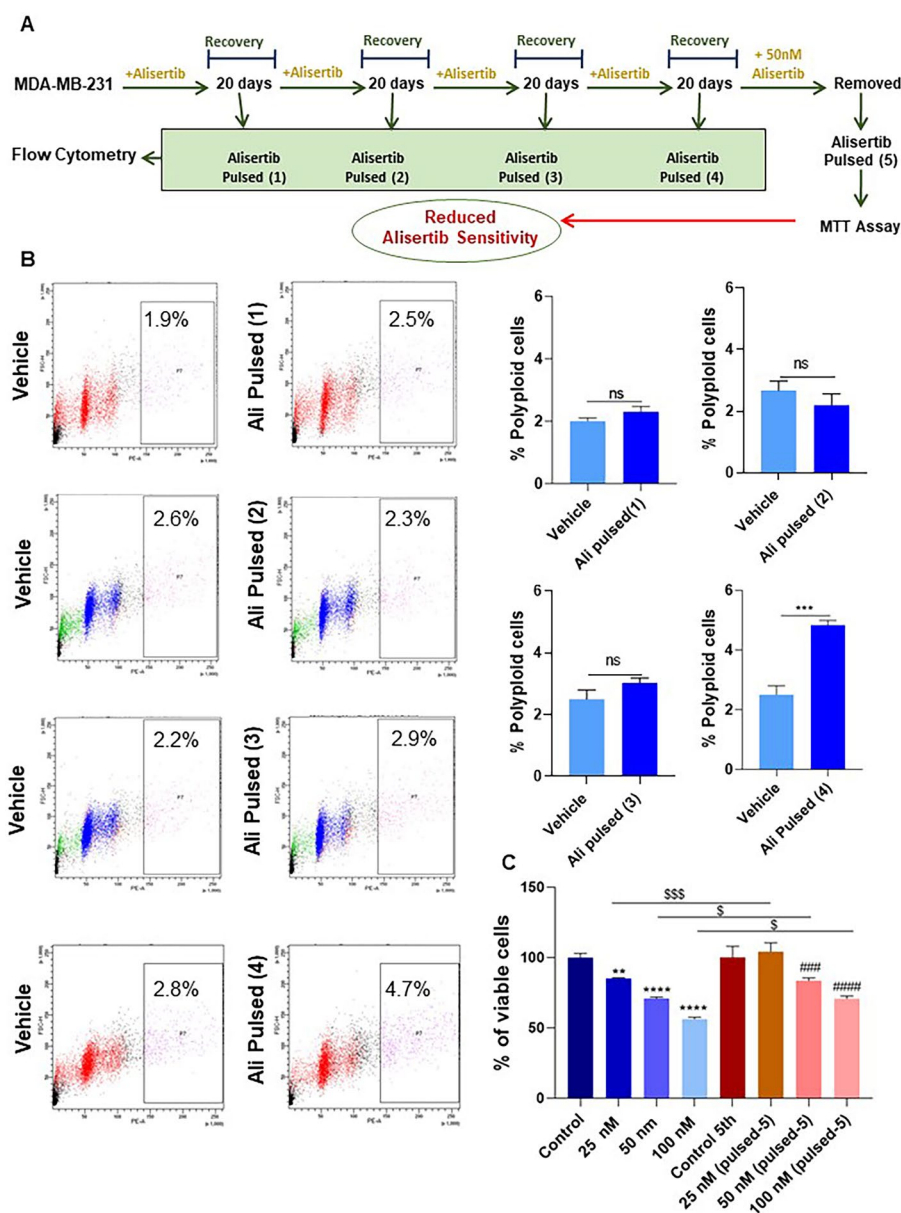


Fig. 8 PGCC-derived daughter cells showed alisertib tolerance upon successive pulses in MDA-MB-231. **A** Schematic study design showing pulsed alisertib administration with subsequent observation of alisertib responsiveness. **B** Flow cytometric dot plots representing the percent population of PGCCs in vehicle vs. pulsed condition. Corresponding column graphs showing polyploid PGCC (%). Experiments were done in triplicates. Values represented mean $\pm$ S.D. ($n=3$). $p<0.05$ was considered as statistically significant. **C** Graphical representation of MTT assay results as done in MDA-MB-231 and pulsed-(5) cells with or without alisertib treatment. The results were represented as (%) of respective controls, and each value represented the mean $\pm$ SD of three independent experiments. The annotations indicated * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$ versus untreated MDA-MB-231 cells. The annotations indicated # $p<0.05$, ## $p<0.01$, ### $p<0.001$ and #### $p<0.0001$ versus untreated pulsed MDA-MB-231 cells. The annotation § indicated a p value <0.05 , §§§ $p<0.001$ as compared between the groups

in tumour derived histocultures. Immunofluorescence imaging conclusively detected expression of GR in PGCCs in both in vitro (Fig. 9A) and histoculture setups (Fig. 9B). Evidential retention of GR among PGCC progenies were also noted (Fig. 9A). We therefore considered mifepristone; a GR antagonist in this study on the basis of earlier reports related to its potential to target PGCC-derived olaparib resistant cells [8] and breast

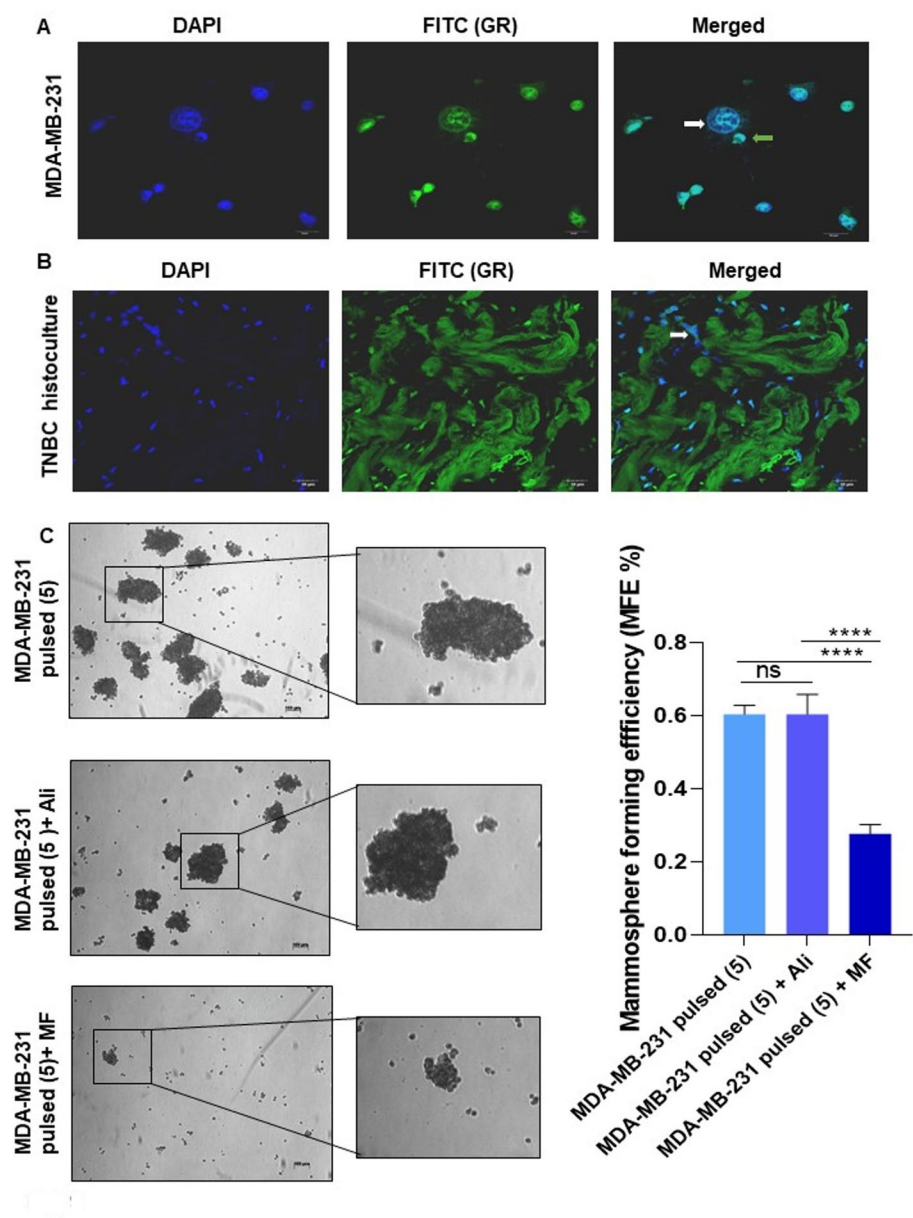


Fig. 9 Mifepristone overcomes alisertib insensitivity in TNBCs. **A** Representative immunofluorescence images of alisertib-treated cells depicting GR expression in PGCCs and their progeny in vitro. White arrow indicated mother PGCC while orange arrow indicated its progeny. Original magnification: 60x. Scale Bar: 20 μ m. **B** Confocal microscopic images of patient-derived histocultures treated with alisertib. The result showed GR expression in PGCCs. Original Magnification: 60x. Scale Bar: 20 μ m. **C** Representative images of formed mammospheres in untreated pulsed-(5) cells (upper row), pulsed-(5) cells treated with alisertib (middle row), pulsed-(5) cells treated with mifepristone (bottom row). Original magnification: 4x. Scale Bar: 10 μ m. Zoomed inset images showing morphological attributes of a single mammosphere formed in each of the above cases (not in scale). Graphical illustration of the relative changes in MFE % was shown in (Right panel)

cancer stem cells [24]. In addition, the compelling analogy between PGCCs and stem like cells in terms of expressional marker profiles along with their ability to fecundate progenies evoked the need for including this anti-fecundity agent in this particular study

Based on these findings, we hypothesized that PGCC-derived alisertib tolerant cells may harbor an enriched pool of alisertib resistant CSCs. For this purpose, we enriched

CSCs by culturing the alisertib pulsed-(5) cells (with or without subsequent alisertib treatment) in non adherent condition by mammosphere forming assay. Results demonstrated that the calculated mammosphere forming efficiency (MFE %) of these cells remain unaltered regardless of alisertib challenge; indicating their remarkable tolerance towards alisertib. However, these tolerant cells when treated with GR antagonist mifepristone, they showed a remarkable drop in mammosphere forming efficiency (Fig. 7C). These results affirmed undoubtedly the potential of mifepristone as an efficient alternative drug to address the cases of alisertib-tolerant TNBCs.

3.10 Mifepristone administration following alisertib restrains PGCC abundance and delays recovery in MDA-MB -231 cells

The remarkable ability of mifepristone to target alisertib-insensitive mammospheres in MDA-MB-231 cells prompted us to investigate the effects of a sequential treatment strategy, where alisertib exposure is followed by mifepristone administration. Phase contrast imaging revealed a visible reduction in polyploid giant cancer cell (PGCC) in response to the sequential alisertib-mifepristone regimen, as illustrated in Fig. 10A (Upper panel). Representative images of morphological characteristics of cells subjected to each condition that is vehicle/alisertib/ alisertib-mifepristone as displayed in the Lower panel showed distinct phenotypic features.

To determine whether this diminished presence of PGCCs, influenced recovery from alisertib-induced stress, we tracked cell populations during a 10-day recovery period. As shown in Fig. 10B, cells treated with alisertib alone exhibited the presence of progeny-producing PGCCs and achieved higher cell density. In contrast, cells subjected to sequential alisertib and mifepristone treatment displayed neither progeny producing PGCCs nor comparable recovery, indicating a significant delay in cellular repopulation.

To further assess whether this impaired recovery was associated with altered clonogenic potential, we performed a colony formation assay. As shown in Fig. 10C cells treated with alisertib followed by mifepristone exhibited a markedly reduced ability to form colonies, in sharp contrast to the alisertib-only group.

These findings reinforce our earlier observations, highlighting the critical role of progeny-producing PGCCs in the recovery mechanism of triple-negative breast cancer (TNBC) cells following alisertib treatment. Targeting these PGCCs with mifepristone appears to effectively disrupt this compensatory survival pathway.

4 Discussion

TNBC patients often suffer from poor prognosis due to absence of ubiquitous availability of agents of targeted therapy. Owing to the important role played by kinases in cell cycle regulation as well as in modulation of cancer hallmark related pathways, several efforts were made to develop effective kinase inhibitors as targeted therapy options. In an attempt to look for targetable kinases with clinical relevance in breast cancer, we thus screened the breast cancer kinome on the basis of CKI scores. Kinases in the T chem category were taken into consideration as they represented the kinases with known biological functions as well as available inhibitors, but yet to receive approval for regular clinical use. For that reason, these kinases may represent the members of the kinome with the highest potential to get included in the routine therapy regimen. Our kinome screening was followed by checking the expression pattern of the top relevant kinases

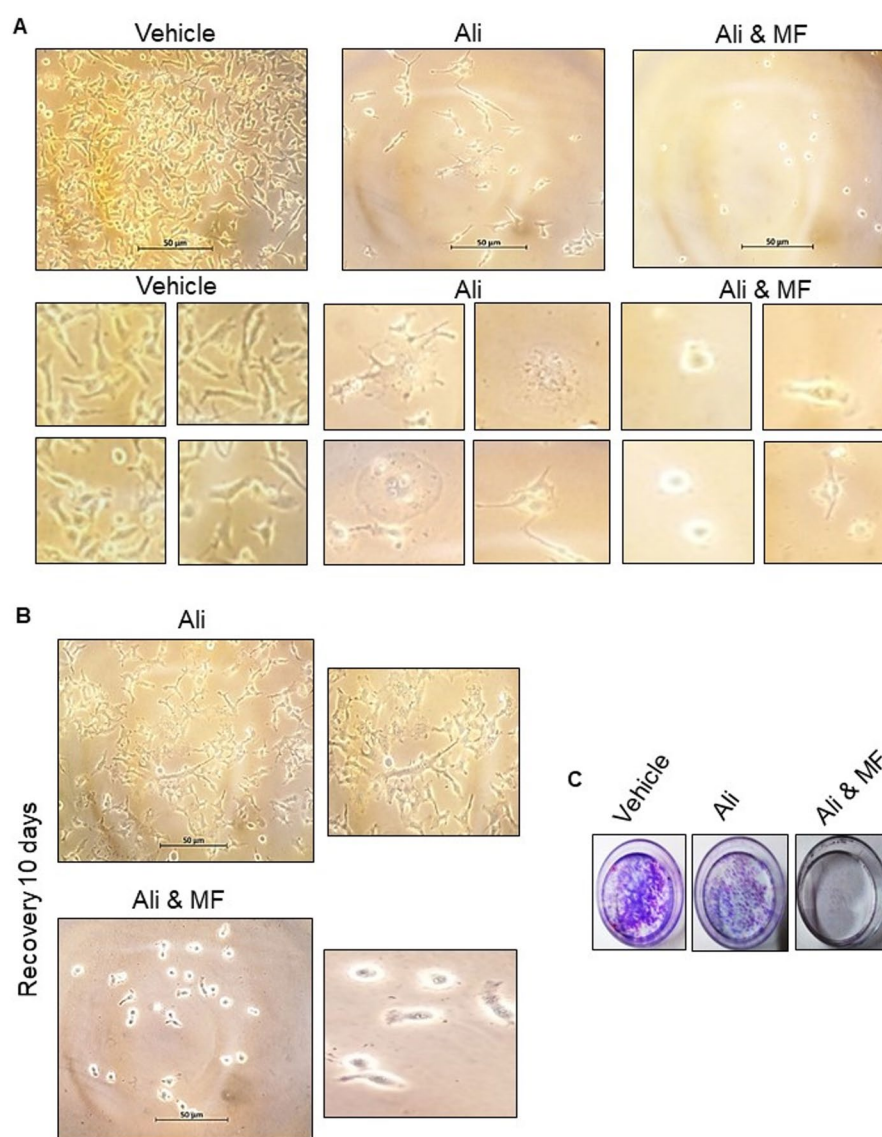


Fig. 10 Mifepristone administration following alisertib treatment reduces PGCC abundance and prolonged recovery in MDA-MB-231. **A** Phase contrast images showing the presence of PGCCs under three different conditions: vehicle control, alisertib treatment, and alisertib followed by mifepristone (Upper panel). Original magnification of the objective: 40X; scale bar: 50 μm . Lower panel highlights morphological features characteristic of each treatment condition. **B** Microscopic images displaying overall cellular density and PGCC abundance following 10 days of recovery from alisertib or sequential alisertib- mifepristone treatment. Original magnification of the objective : 40X. Scale bar : 50 μm . **C** Colony formation assay demonstrating reduced clonogenic potential in cells treated with successive alisertib-mifepristone treatment, compared to cells treated with alisertib alone

in TCGA mentioned cancer types, with the notion that those kinases with pan overexpression in all cancer types may offer a significant impact on the progression of cancer hallmarks. However, not overexpression but the ultimate prognostic effect signifies the clinical value of a kinase. Thus, kinases, which are overexpressed in 50% of TCGA cancer types, were evaluated for their effect on overall survival of breast cancer patients. The results displayed the unequivocal significance of cell cycle regulator Aurora kinase A in breast cancer that was also overexpressed in our patient cohort as well as in TNBC

patients screened from publicly available datasets. These findings prompted to evaluate the potential of Aurora kinase A to become an effective drug target, which shows its druggability potential.

As our result portrayed the prospect of Aurora kinase A as a druggable kinase in breast cancer, we therefore searched for available drugs to target the same. Drug interaction screening led us to alisertib, the only Aurora kinase A inhibitor to reach phase 3 clinical trial [25]. Alisertib was thus compared with routine chemotherapeutics for an assessment of pharmacokinetic, physicochemical properties, bioavailability, and organ toxicity. Our *in silico* result displayed satisfactory performance of alisertib, thereby upholding its acceptability as a potent tolerable drug targeting Aurora kinase A. However, results from earlier clinical trial unexpectedly showed that alisertib although performed well in non-TNBCs, it failed to offer added benefit in TNBC patients [7]. Concerning the adequately acceptable properties of alisertib together with the procedural complications in developing new alternative drugs, it was felt to explore the underlying causes of alisertib insensitivity in TNBCs. This seemed to be the best viable option with utmost necessity. Interestingly, Aurora kinase A inhibition is reported to favour formation of polyploid giant cancer cells that resist sensitivity to anti-mitotic drugs [26]. Thus, we explored this possibility in our setup considering inhibition of Aurora kinase A signaling by alisertib. An enrichment of PGCCs were indeed evident in a dose dependent manner subject to alisertib treatment. These PGCCs retained replicative ability, possibly undergoing multiple rounds of replication and ultimately produced progeny cells during drug holiday. Our results concluded that single PGCCs could fecundate the initiation of tumor cell colonies. Such characteristics of PGCCs indeed indicated their stem-like abilities that is demonstrated by retention of key stemness and EMT markers in such PGCCs and also their progeny irrespective of alisertib treatment. Our *in vitro* results thus pointed out generation of progeny from alisertib insensitive PGCCs, which hold potential to overcome alisertib-induced stress.

However, *in vitro* setups alone, only partially replicate the complexity of patient scenario due to a lack of the complicated cell-stroma interactions and cellular heterogeneity that defines the complex *in vivo* condition and dictates treatment response [27]. Patient-derived primary culture models are also hard to maintain and require the use of enzymatic dissociation procedures that yet again disrupt the complex tissue architecture of the tumor microenvironment which may alter the response towards a tested drug [27]. Patient derived organoids and xenografts also do not maintain the original intricacies of human tissue. On the other hand, histocultures, as they can better preserve the original tumor architecture; more closely represented the patient conditions than their 2D *in vitro* counterpart or patient derived primary cultures. Moreover, histocultures demonstrated remarkable efficacy in effectively predicting responses to drugs in several cancer models that closely mimicked findings of clinical trials thus validating its importance as a successful preclinical model [27–29]. Thus, we undertook a novel approach of testing the effects of alisertib in a TNBC tumour derived histoculture explant model. Our findings in treated tissue explants re-supported our *in vitro* observation of PGCC enrichment and further exhibited expression of key EMT, stemness proteins in these cells. With the support from the explant culture findings, it was concluded that even in the patient scenario, alisertib boosted PGCC percentage.

PGCCs are reported to be important players in acquisition of therapy resistance [30]. Thus, this study focused further on the consequence of such PGCC enrichment. A model consisting of repeated alisertib pulses followed by drug holidays was thus initiated. A persistent

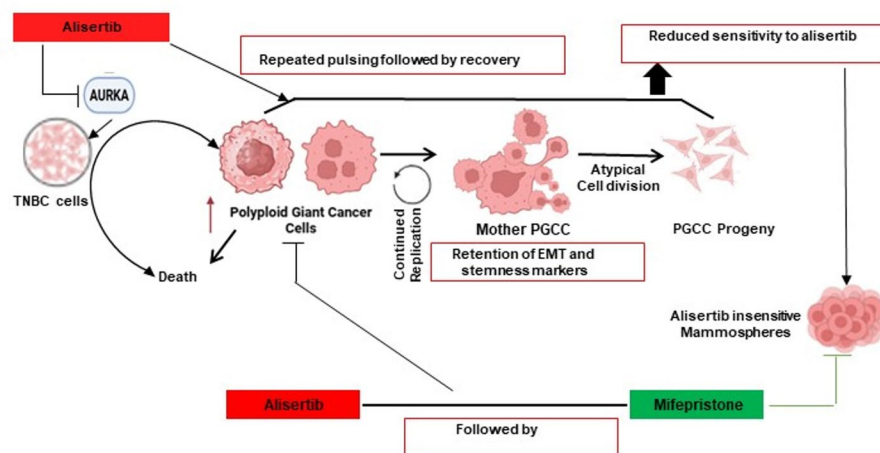


Fig. 11 Schematic representation of findings related to the mechanism of alisertib insensitivity in TNBCs and its redressal by mifepristone. Images were created with BioRender.com

PGCC enrichment after repeated pulses was indeed indicative of an adaptive change that ultimately culminated into development of reduced alisertib sensitivity; presumably from the fecundated progenies. Therefore, this alisertib insensitivity in TNBCs following PGCC enrichment affirmed our hypothesis that such insensitivity does stem from PGCCs.

As our previous results denoted single cell clonogenicity and stem like features in PGCCs in line with previous reports of acquirement of embryonic stemness by PGCCs [31], we assumed that alisertib insensitivity may be a consequence of enrichment of stem like cells that remain insensitive to alisertib. Thus, CSC population in these cells were enriched by mammosphere formation. Mammospheres were thereafter targeted with alisertib that expectedly failed to evoke any satisfactory response. This further compelled us to look for alternative agents that may target these alisertib tolerant cells. A recent report demonstrated the remarkable ability of mifepristone, an anti-fecundity agent that also acts as GR antagonist in reverting PGCC-derived acquired resistance to olaparib in ovarian cancer; a possible consequence of altered GR signaling [8]. GR is intricately involved in chemoresistant CSC enrichment [12], metastatic progression [32] and thus leads to poor prognosis in TNBCs [11]. GR status in PGCCs was thus examined which showed its expressional persistence in mother PGCCs as well as in daughter cells. Mifepristone's potential was thus tested in the alisertib-insensitive mammospheres. Mifepristone showed remarkable efficacy in targeting this resistant population as marked by low mammosphere forming efficiency of such cells. Moreover, a sequential alisertib-mifepristone therapy effectively reduced PGCC abundance and delayed recovery and clonogenicity in comparison to alisertib monotherapy. Therefore, mifepristone may be an effective drug targeting alisertib insensitive cells in TNBCs. The overall findings related to this study has been elucidated schematically in Fig. 11.

5 Conclusion

This study shed some light on the determinant role of PGCCs and their daughters in dictating responsiveness towards Aurora kinase A inhibitor alisertib in TNBCs. Therefore, understanding the underlying mechanism of PGCC-derived alisertib insensitivity may aid to develop better strategies to counteract the same. The study addressed this issue with the

GR antagonist, mifepristone which is also known for its anti-fecundity property. Mifepristone effectively targeted alisertib insensitive BCSCs and delayed recovery during sequential alisertib-mifepristone treatment. These overall findings may pave the path towards future research on use of successive combinatorial approach wherein alisertib insult may be substantiated by successive administration of mifepristone pulses. This might offer better prognosis in TNBCs with a target-directed approach.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s12672-025-03825-0>.

Supplementary material 1.

Supplementary material 2.

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Author contributions

DT: performed the entire experiments, acquired and analysed the data and wrote the manuscript; DS: lead the team in performing experiments; SK: literature search and integrated some of the obtained experimental data; JC: overall supervision of the study with clinical advice; SS: helped in collecting surgical tissues along with medical advice; SH: histopathological examination and identification of tumour and adjacent normal; AL, EM and SD: participated in performing some of the experiments; PK: senior advisory; SM: designing of the study, supervision, editing, checking and final approval of the manuscript. All authors reviewed the manuscript.

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Data availability

Publicly available datasets with accession numbers GSE21422 and GSE61304 were analyzed in this study. This data can be found at: <https://www.ncbi.nlm.nih.gov/gds/>. All other data are available from corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The Institutional Ethics Committee of Chittaranjan National Cancer Institute, Kolkata, INDIA that follows the guidelines of Central Drugs Standard Control Organisation under the Directorate General of Health Services, Ministry of Health & Family Welfare, Government of India has approved this study with Ethics approval number # CNCI-IEC-SM-20, dt 28.7.2022. Surgically removed tissue samples were collected after signing of Informed Consent Form in accordance with the Helsinki Declaration.

Consent for publication

Consent for publication was obtained from all the individuals.

Competing interests

The authors declare no competing interests.

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